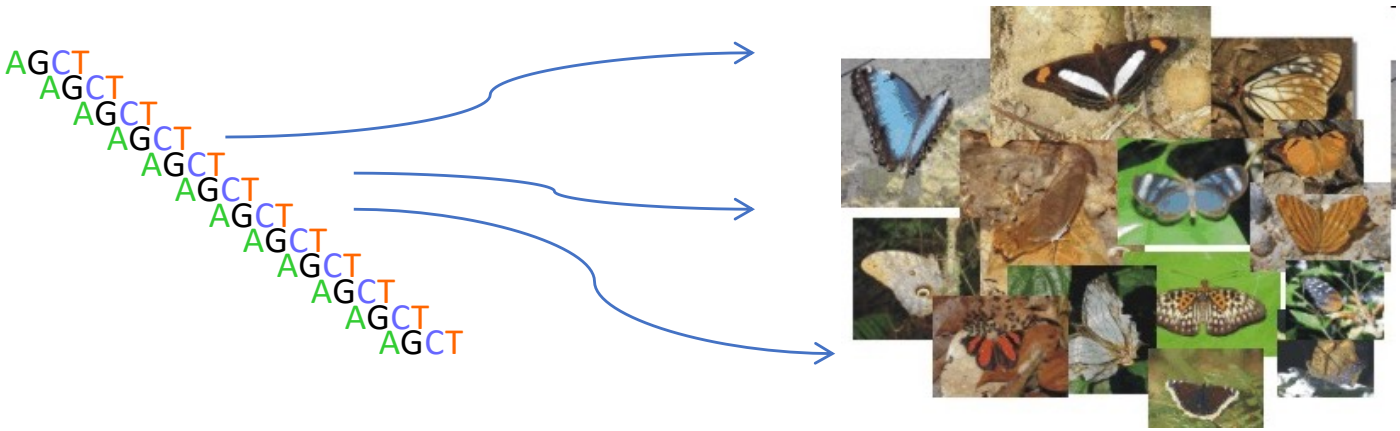
BIOR90 Evolutionary Biology - Methods and Applications 2026

Teachers Jadranka Rota, Niklas Wahlberg, Etka Yapar, Emma Kärrnäs

Recap: Why *molecular* systematics?

- Ease of data generation for large numbers of taxa
- Ease of generating a large number of independent data sets for given taxa
- Molecular characters behind the morphological characters we see



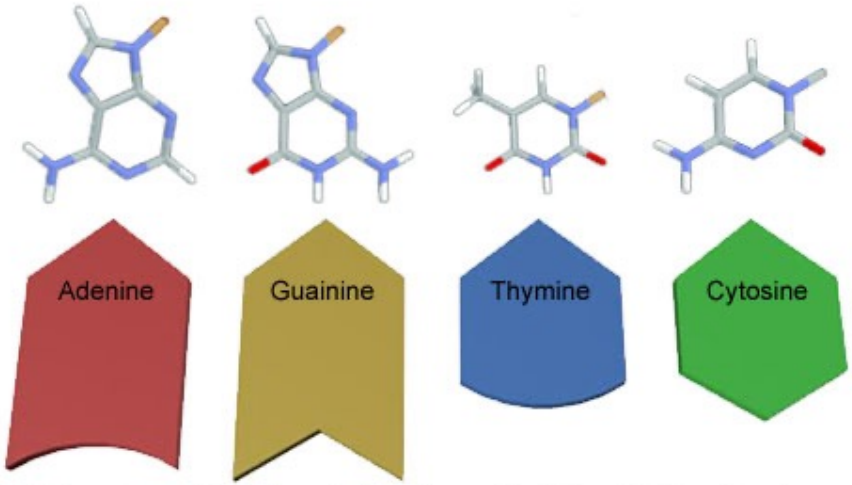
DNA as a source of information

- **DNA has four characters**

Purines

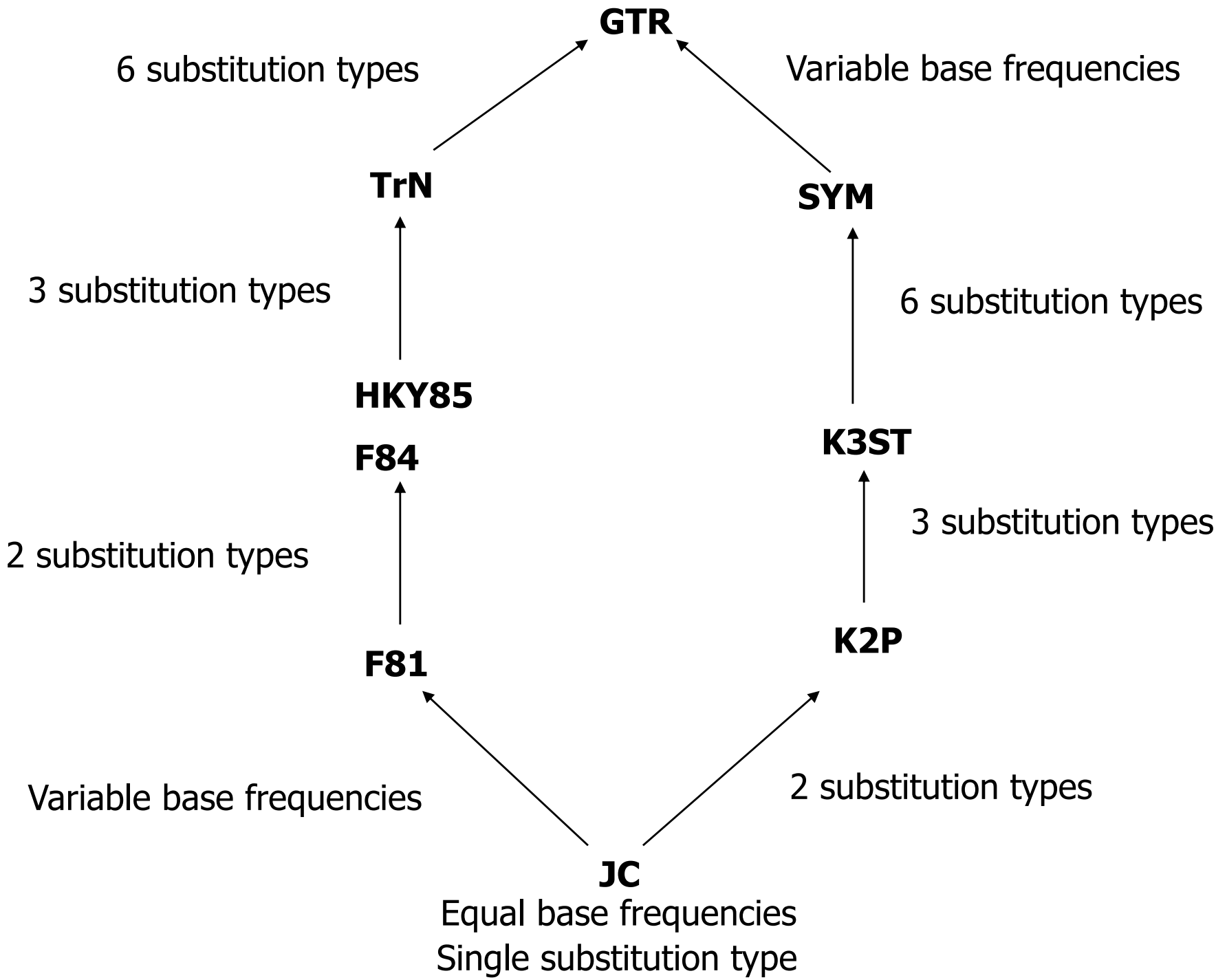
Pyrimidines

Figure B-3: The Four Nitrogenous Bases



Each base has a distinct shape that can be used to distinguish it form the others. 3D representations of the four bases are shown, with the corresponding chemical structures drawn above.

DNA Evolution Models



Model-based approaches:

Maximum Likelihood

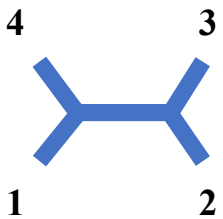
- For reconstructing phylogenies

Model

  - which tree topology (τ), branch lengths, and parameters of DNA evolution model (θ) (e.g. transition/transversion ratio, base frequencies, …) are maximizing the probability of observing the sequences at hand?

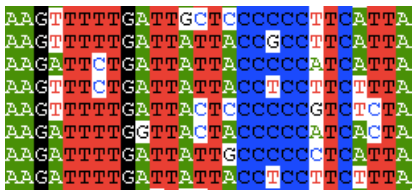
Data

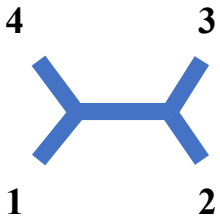
$$L(\tau,\theta) = \Pr(\text{Data} \mid \tau,\theta)$$



	A	C	G	T
A		α	α	α
C	α		α	α
G	α	α		α
T	α	α	α	

=





	A	C	G	T
A		α	α	α
C	α		α	α
G	α	α		α
T	α	α	α	

ML analysis in short

- Tree topology is obtained
- Branch lengths and parameters of the DNA substitution model are optimized
- Different topologies (with branch lengths and DNA substitution model parameters optimized) are compared based on their likelihood as the optimality criterion
- The topology with the highest likelihood needs to be found

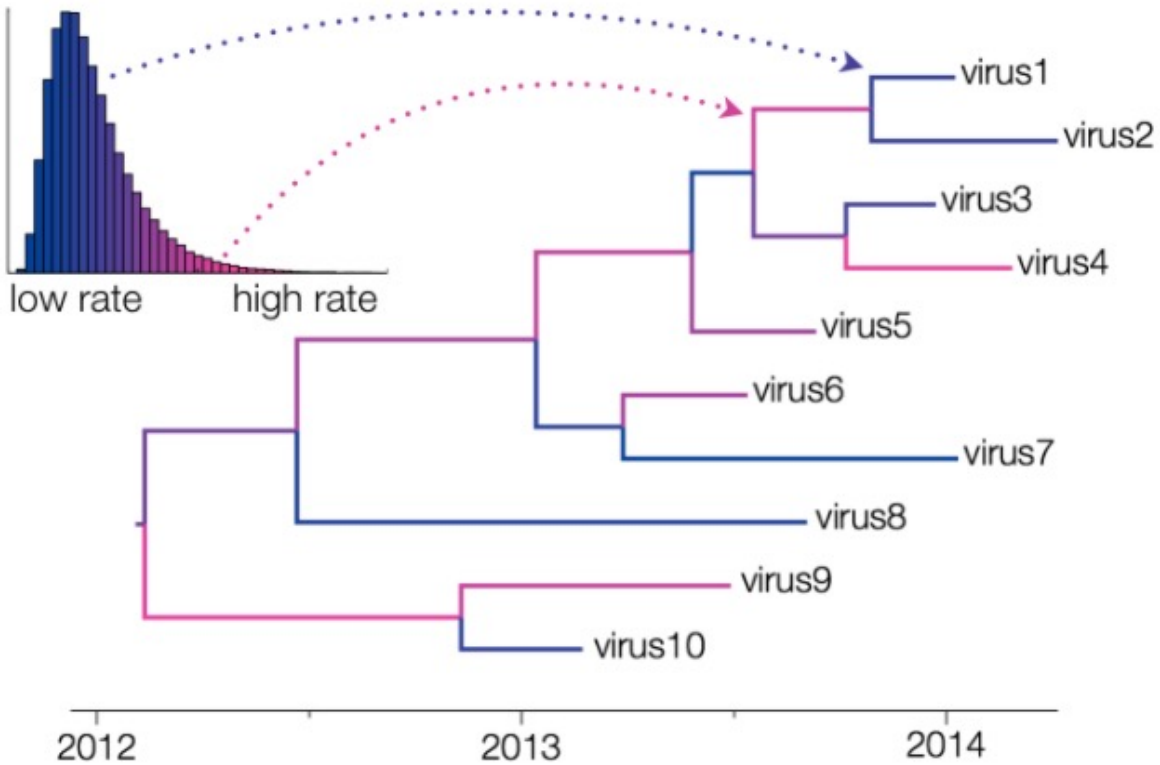
A Bayesian approach compared to ML

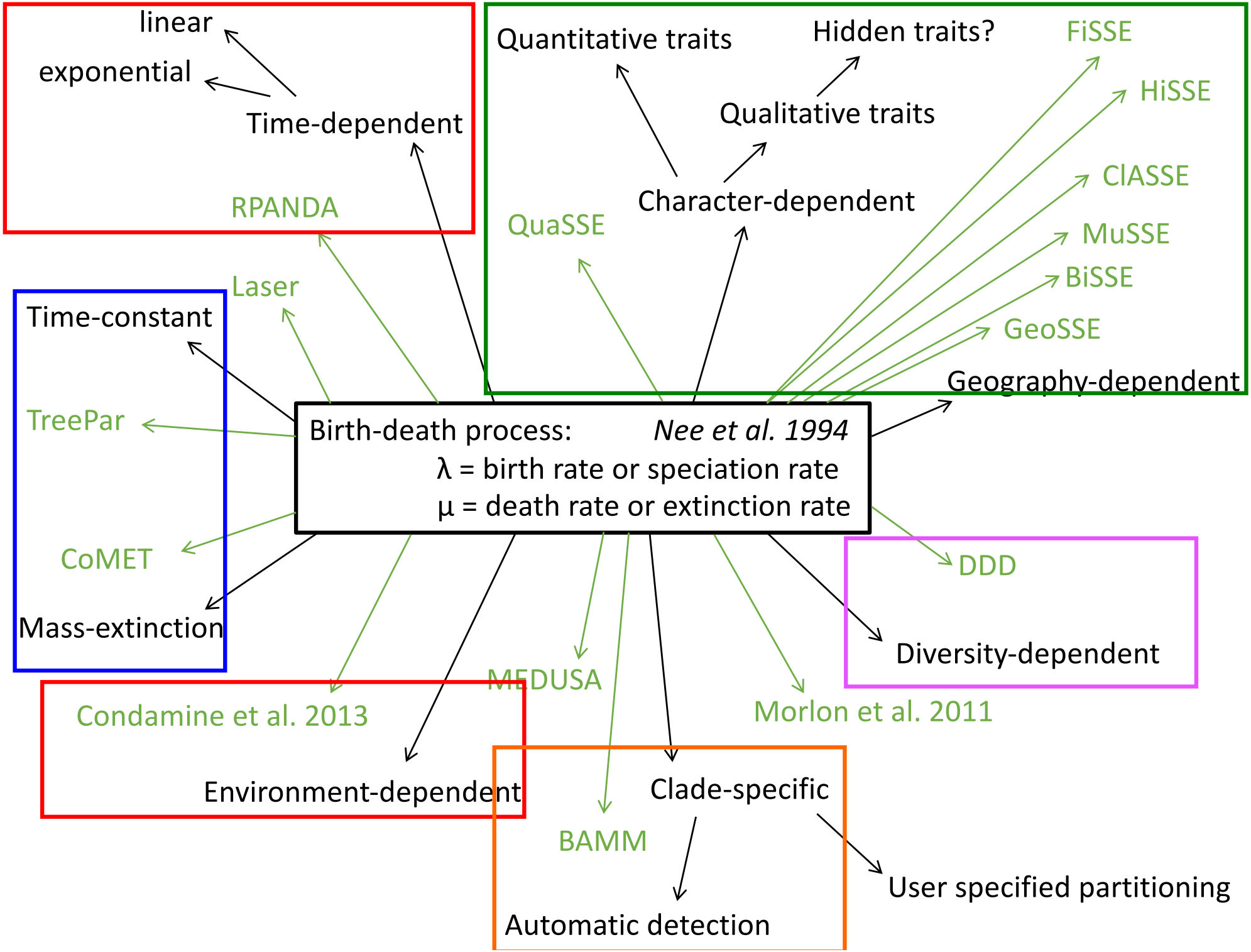
- The likelihood is the probability of observing the data given a hypothesis
  - $L = Pr(D \mid \boldsymbol{\theta})$.
- **In ML** we search for the parameter values of the model that maximize the likelihood function

- **In a Bayesian analysis, we get the probability of a hypothesis given the data (probability of the tree given the sequences)**
  - We combine the likelihood of a given hypothesis with a prior expectation for this hypothesis to obtain a posterior probability of the hypothesis

Uncorrelated relaxed clocks

- Models available in *BEAST*
  - **Lognormal distribution**
    Most rates cluster around the mean
  - **Exponential distribution**
    Most rates are quite low



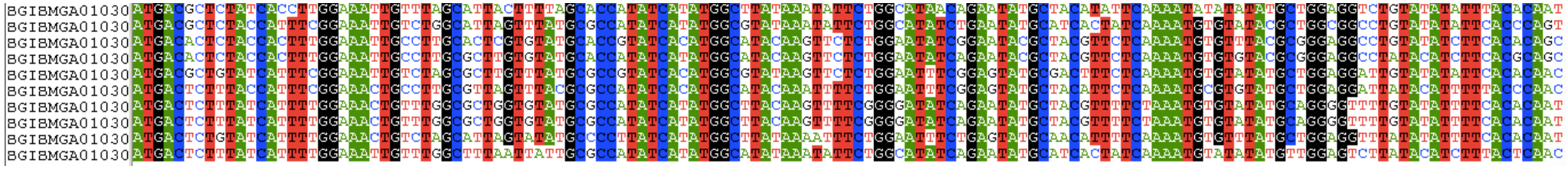

linear
exponential
Time-dependent
RPANDA
Quantitative traits
Hidden traits?
FiSSE
HiSSE
Qualitative traits
ClASSE
Character-dependent
QuaSSE
MuSSE
BiSSE
Laser
Time-constant
GeoSSE
Geography-dependent
TreePar
Birth-death process: Nee et al. 1994
λ = birth rate or speciation rate
μ = death rate or extinction rate
CoMET
DDD
Mass-extinction
Diversity-dependent
MEDUSA
Condamine et al. 2013
Morlon et al. 2011
Environment-dependent
Clade-specific
BAMM
User specified partitioning
Automatic detection

This week in BIOR90 – how to analyse data

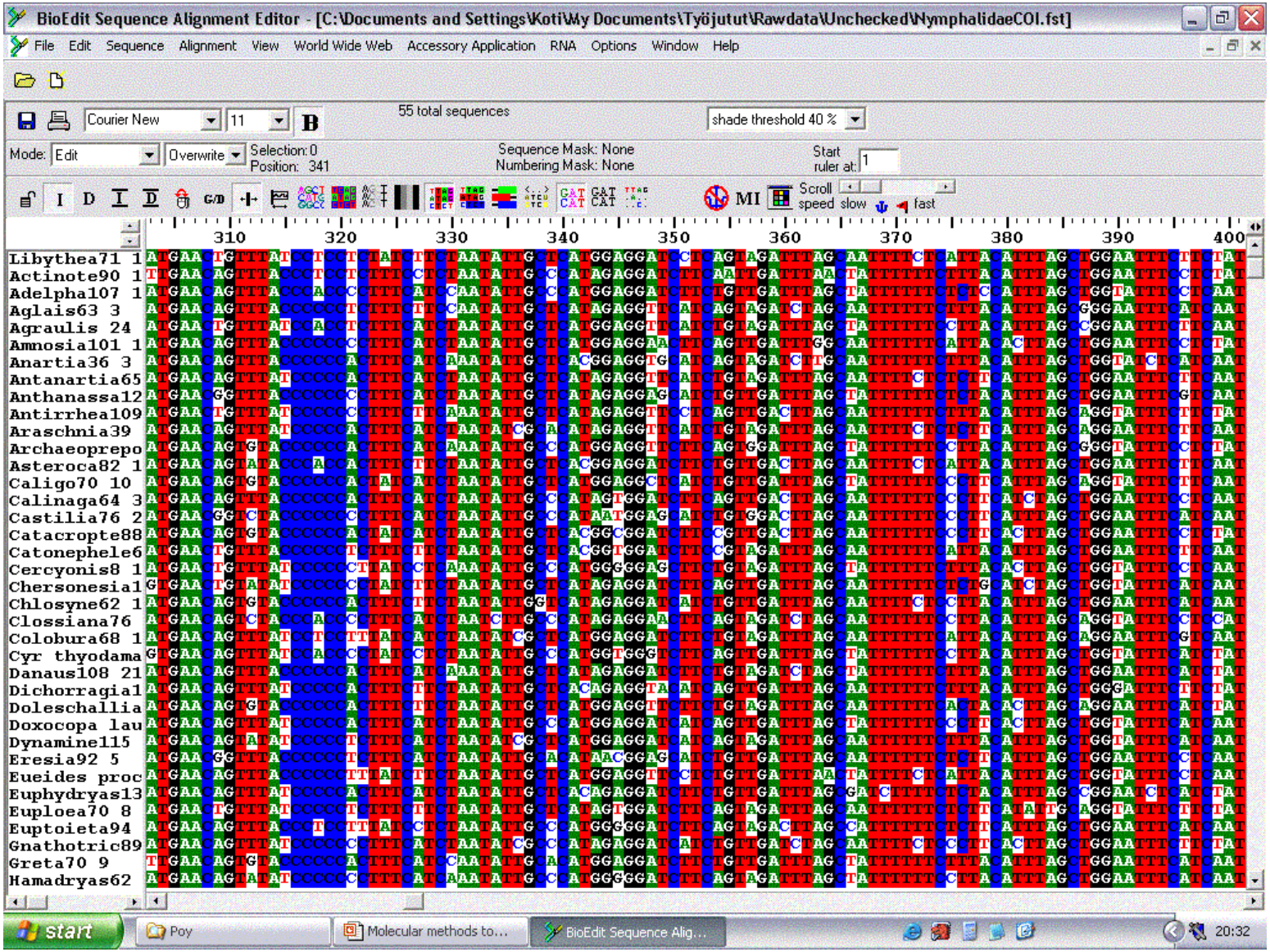
Hours\Days	Wed Apr 29	Mon May 4	Tue May 5	Wed May 6
9:00-10:00	Module outline by Charlie Cornwallis	Tutorials 3-4 – models, ML (EY, JR)	Tutorial 8 – diversification (EY, NW)	Tutorial 9 – gene tree/species tree (EK, EY)
10:00-12:00	Introduction to alignments, file formats (NW)	Tutorial 5 – Bayesian (EY, JR)	Tutorial 8 – diversification (cont.) (EY, NW)	Tutorial 10 – historical biogeo. (EK, EY, JR)
12:00-13:00	Lunch	Lunch	Lunch	Lunch
13:00-14:30	Tutorial 1 – creating datasets (EY, JR)	Tutorial 6 – timing of divergence (EY, NW)	Memorial seminar Bengt Olle Bengtsson	Catch up time (EY, JR)
14:30-16:00	Tutorial 2 – alignment (EY, JR)	Tutorial 7 – tree manipulation (EY, NW)	Memorial seminar Bengt Olle Bengtsson	Catch up time (EY, JR)

Tutorials on: **https://github.com/NymphalidNiklas/EB2_2026**

Multiple Sequence Alignment



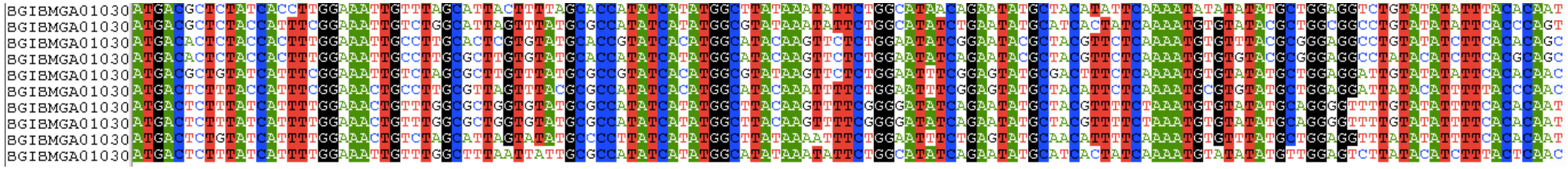
Alignment can be easy…



...or difficult

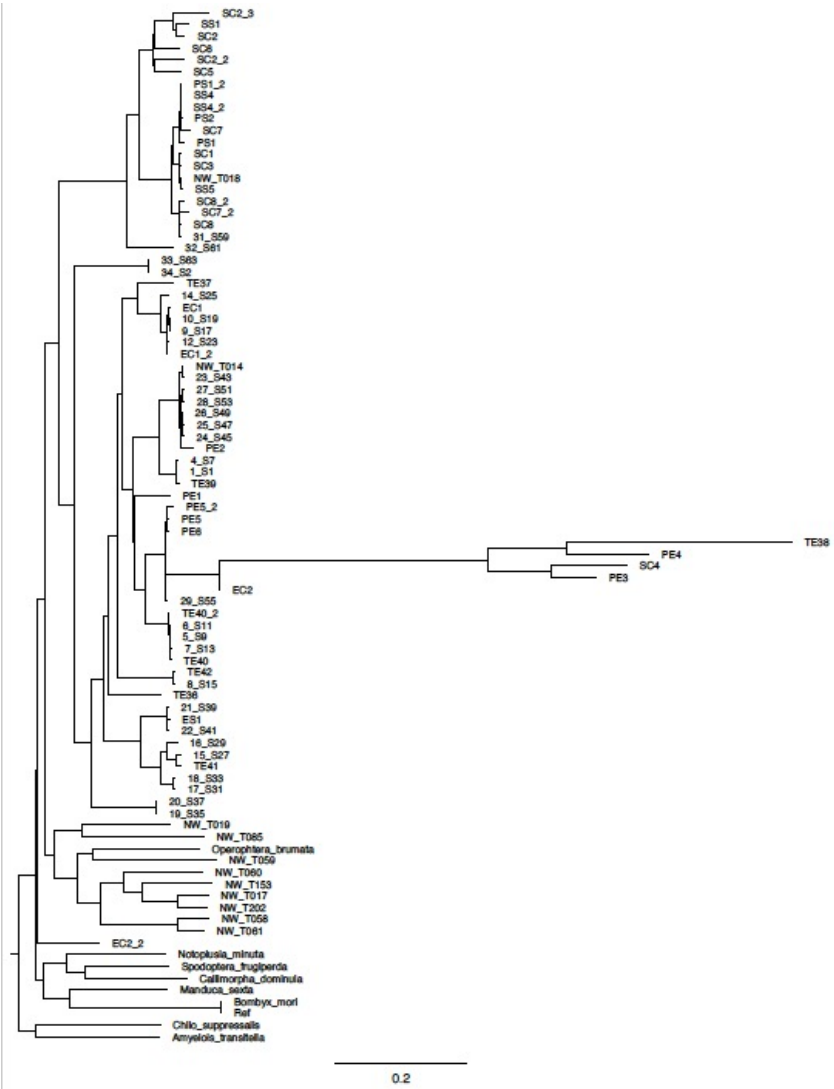
Homology: Definition

- Homology: similarity that is the result of inheritance from a common ancestor - identification and analysis of homologies is central to phylogenetic systematics
- An alignment is a hypothesis of positional homology between bases/amino acids



Multiple sequence alignment- goals

- To generate a concise, information-rich summary of sequence data
- Alignments can be treated as models that can be used to test hypotheses
- Does this model of events accurately reflect known biological evidence?



Multiple sequence alignment

- Manual
- Dynamic programming
- Heuristic methods
  - Progressive alignment
  - Consistency-based scoring
  - Iterative refinement methods

Manual alignment - reasons

- Might be carried out because:
- Alignment is easy
- There is some extraneous information (structural)
- Automated alignment methods have encountered a local minimum problem
- An automated alignment method can be “improved”

Protein-coding genes can often be manually aligned

How to align these sequences:

AGGGCTTTAA
AGGCTA
AATGGCTCTAA
GGAGCCCTAA

How to align these sequences:

```
A-GGGCTTTAA
A--GGCT--A-
AATGGCTCTAA
GGAG-CCCTAA
```

How to align these sequences:

```
-AGGGCTTTAA
-A-GGC--TA-
AATGGCTCTAA
-GGAGCCCTAA
```

Multiple sequence alignment

- Is not easy! How to be objective?
- Dynamic programming
  - Works only for a very small number of sequences
  - Doesn’t use phylogeny and doesn’t incorporate an evolutionary model
- Heuristic methods
  - Progressive alignment
    - E.g. Clustal
  - Consistency-based scoring
    - Used in e.g. T-Coffee
  - Iterative refinement methods
    - Used in Mafft

Heuristics: Progressive alignment

- Devised by Feng and Doolittle in 1987
- A heuristic method and as such is not guaranteed to find the ‘optimal’ alignment
- Most successful implementation is Clustal
  - ClustalW
  - ClustalX

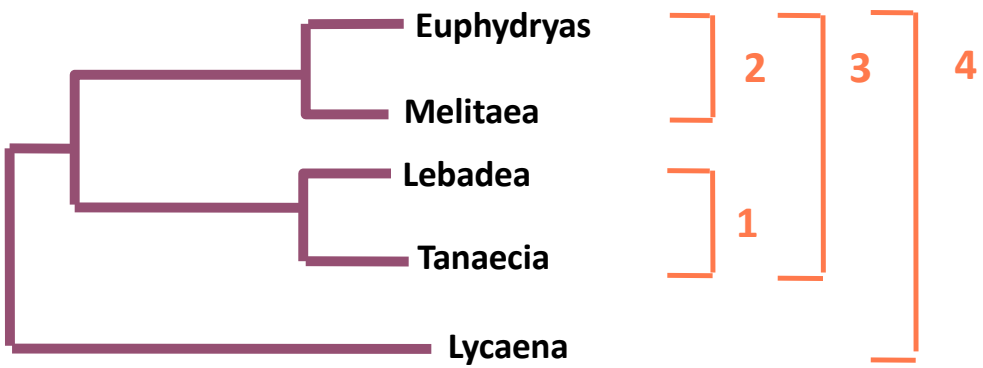
Clustal - pairwise alignments

- First perform all possible pairwise alignments between each pair of sequences
- Calculate the 'distance' between each pair of sequences based on these isolated pairwise alignments
- Generate a distance matrix

Taxon	*Euphydryas*	*Melitaea*	*Lebadea*	*Tanaecia*	*Lycaena*
Euphydryas	-				
Melitaea	0.17	-			
Lebadea	0.59	0.60	-		
Tanaecia	0.59	0.59	0.13	-	
Lycaena	0.77	0.77	0.75	0.75	-

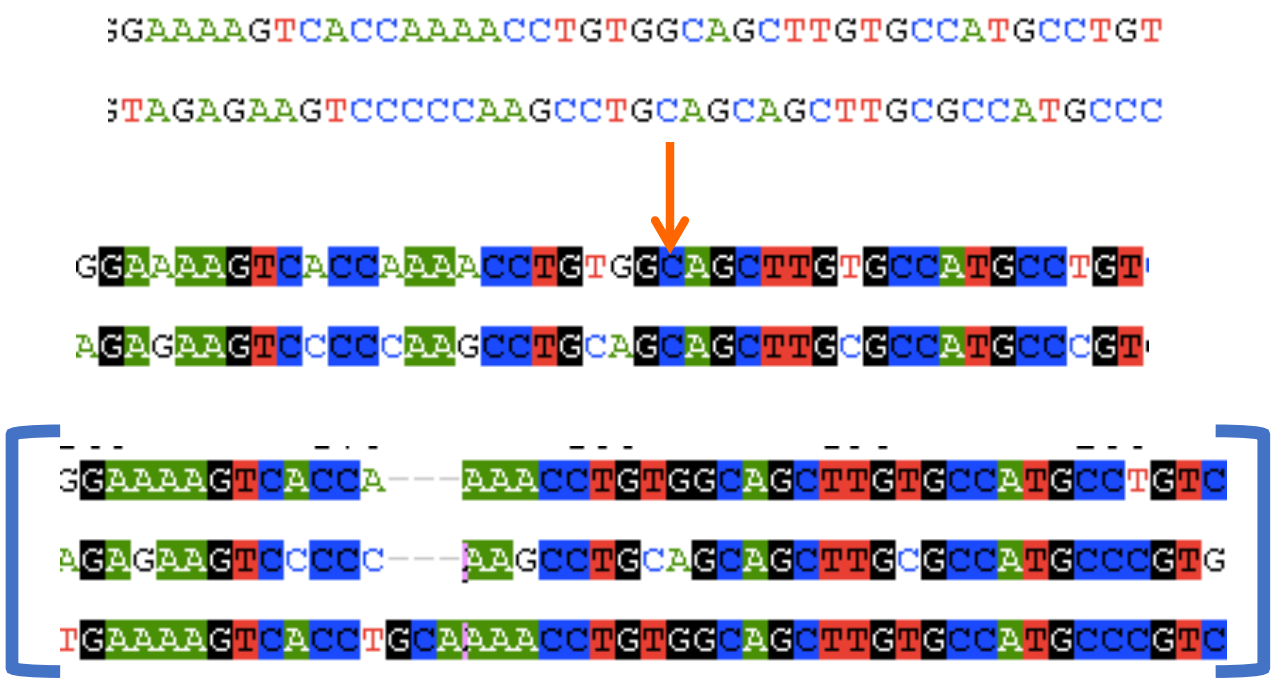
Clustal - guide tree

- Generate a Neighbour-Joining 'guide tree' from these pairwise distances
- This guide tree gives the order in which the progressive alignment will be carried out



Multiple alignment- first pair

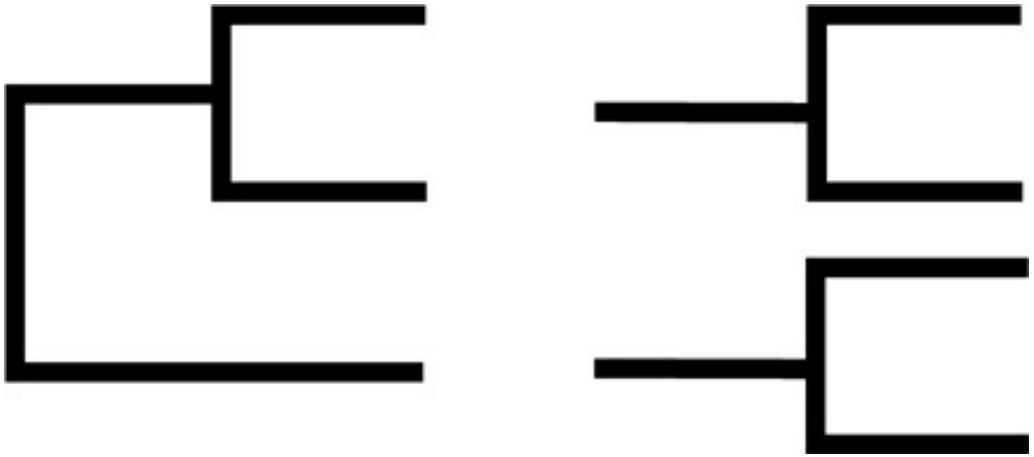
- Align the two most closely-related sequences first
- This alignment is then ‘fixed’ and will never change
- If a gap is to be introduced subsequently, then it will be introduced in the same place in both sequences, but their relative alignment remains unchanged



Clustal - decision time

- Consult the guide tree to see what alignment is performed next.
  - Align a third sequence to the first two

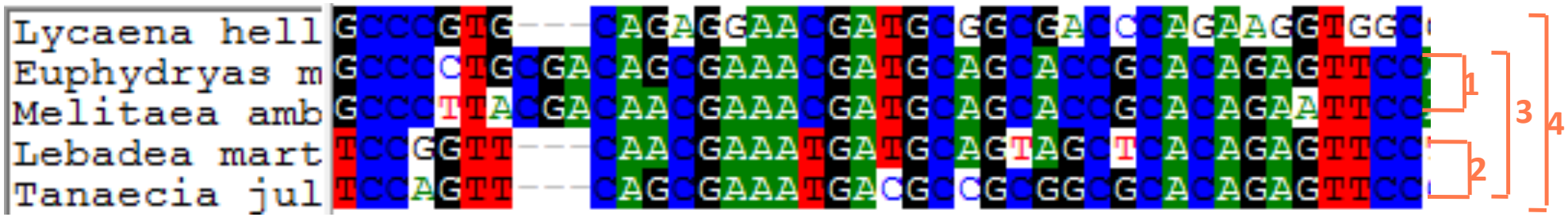
  Or
  - Align two entirely different sequences to each other.



Option 1 **Option 2**

Clustal - progression

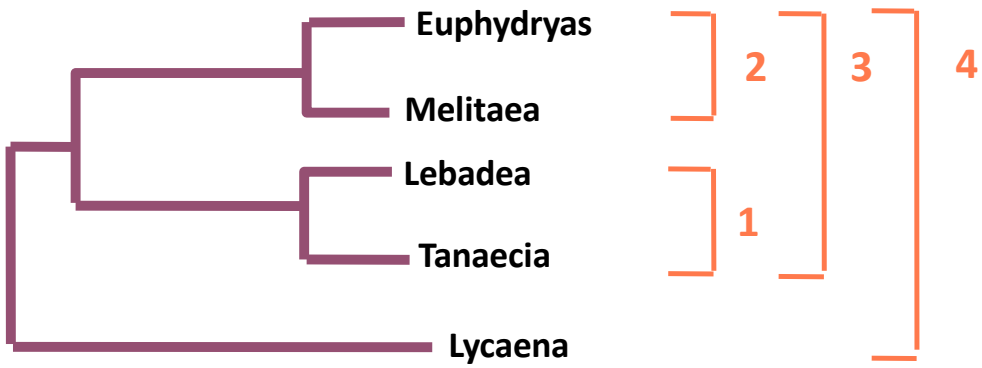
- The alignment is progressively built up in this way, with each step being treated as a pairwise alignment, sometimes with each member of a ‘pair’ having more than one sequence



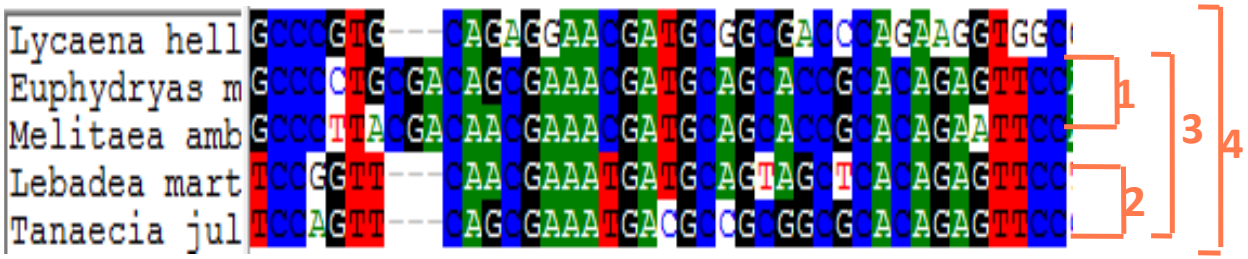
Overview of Clustal procedure

Taxon	*Euphydryas*	*Melitaea*	*Lebadea*	*Tanaecia*	*Lycaena*
Euphydryas	-				
Melitaea	0.17	-			
Lebadea	0.59	0.60	-		
Tanaecia	0.59	0.59	0.13	-	
Lycaena	0.77	0.77	0.75	0.75	-

Quick pairwise alignment: calculate distance matrix



Neighbour-joining tree (guide tree)



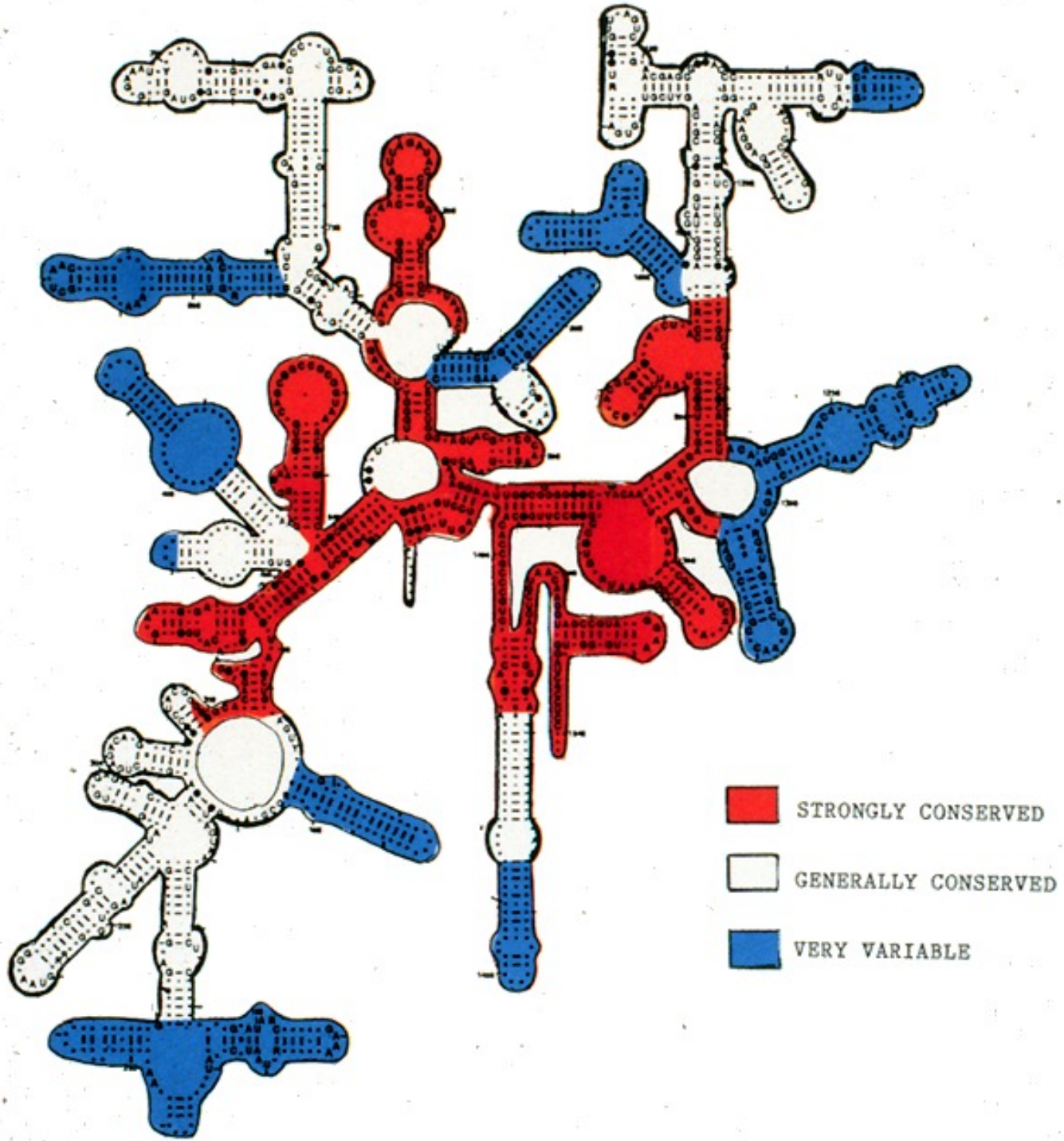
Progressive alignment following guide tree

Clustal - good points/bad points

- Advantages:
  - Speed
- Disadvantages:
  - Hierarchic structure introduced that is not necessarily phylogenetic
  - No way of quantifying whether or not the alignment is good
  - No way of knowing if the alignment is ‘correct’
  - Local minimum problem. If an error is introduced early in the alignment process, it is impossible to correct this later in the procedure
  - Arbitrary alignment

Increasing the sophistication of the alignment process

- Should we treat all the sequences in the same way?
  - some sequences are closely related and some sequences are distant relatives.
- Should we treat all positions in the sequences as though they were the same?
  - they might have different functions and different locations in the 3-dimensional structure.
  - codon structure – how to retain this?


STRONGLY CONSERVED
GENERALLY CONSERVED
VERY VARIABLE

Iterative refinement methods

- Initial alignments split into two groups randomly
- Within groups the alignment is kept fixed
- Dynamic programming used to align the two groups to each other
- This is repeated until score converges
- Used in e.g. Muscle and MAFFT

Using models in alignment

- New methods are being developed all the time
- Latest methods include using a Bayesian statistic framework, DNA evolutionary models and alignment concomitantly with estimation of phylogentic relationships
- Still not feasible with a moderately sized dataset

Bottom line

- Alignments are extremely important in phylogenetics
- A bad alignment means many wrong statements of homology, which means pure rubbish as output
- A good alignment can be hard to attain

The Tree

Finding the optimal trees

Numbers of possible trees for N taxa

1	1
2	1
3	1
4	3
5	15
6	105
7	945
8	10395
9	135135
10	2027025
11	34459425
12	654729075
13	13749310575
14	316234143225
15	[illegible]
[illegible]	[illegible]4766997 71875 (2 x 10^{20})
50	3 x 10^{74}

How can we find the most optimal tree?

Astronomers estimate that the universe could contain **up to one septillion stars** – that's a one followed by 24 zeros. Our Milky Way alone contains more than 100 billion, including our most well-studied star, the Sun.

NASA

Finding optimal trees - exact solutions

- Exact solutions can only be used for small numbers of taxa
- Exhaustive search examines all possible trees
- Branch and bound does not examine all trees, but will find optimal tree(s)
- Typically used for problems with 10–20 taxa

Finding optimal trees - heuristics

- The number of possible trees increases faster than exponentially with the number of taxa making exhaustive searches impractical for many data sets (an NP-complete problem)
- Heuristic methods are used to search tree space for optimal trees by building or selecting an initial tree and swapping branches to search for better ones
- The trees found are not guaranteed to be optimal - they are best guesses

Finding optimal trees - heuristics

- Stepwise addition

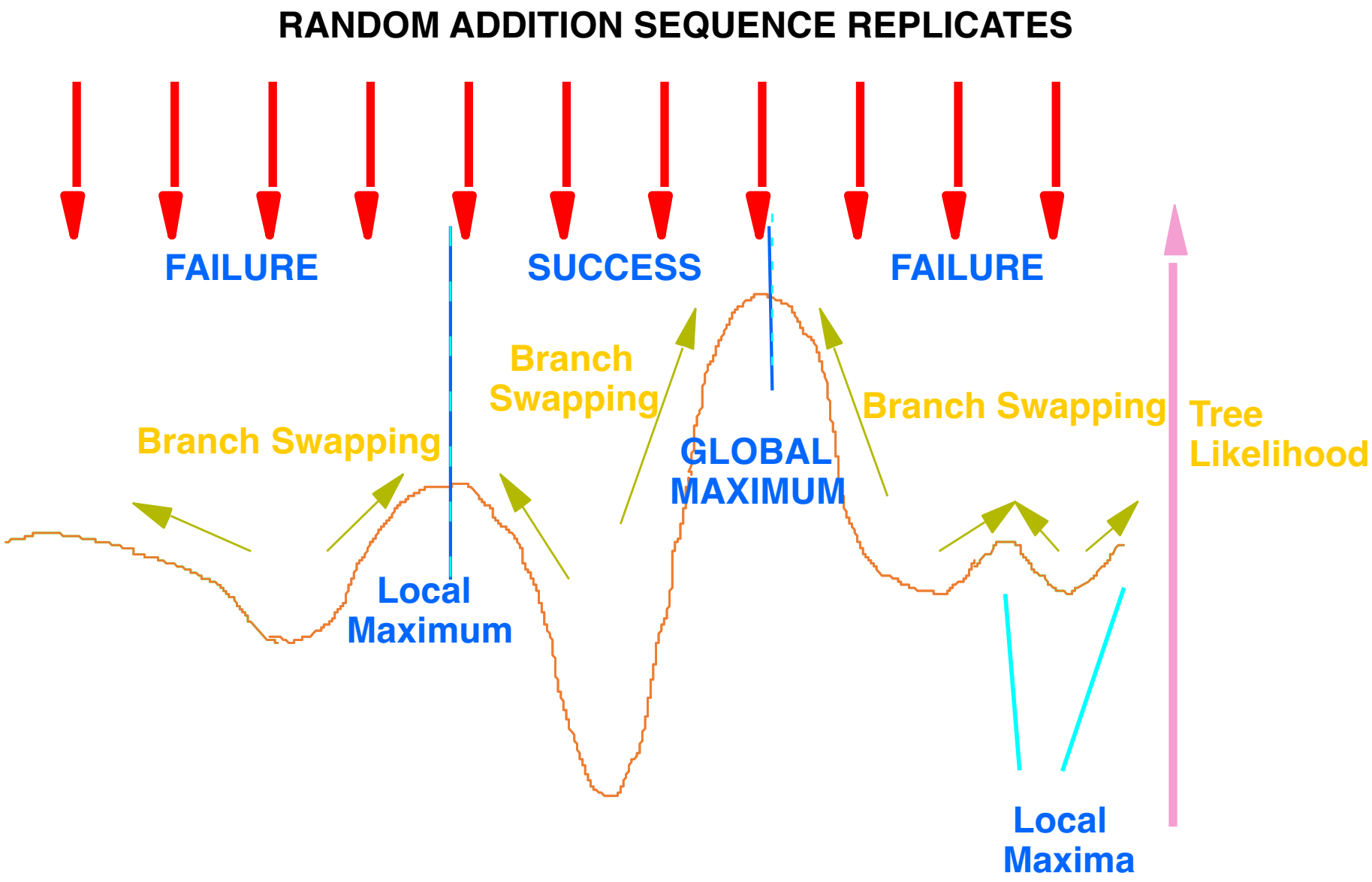
Asis - the order in the data matrix

Closest - starts with shortest 3-taxon tree, adds taxa in order that produces the least increase in tree length (greedy heuristic)

Simple - the first taxon in the matrix is taken as a reference - taxa are added to it in the order of their decreasing similarity to the reference

Random - taxa are added in a random sequence, many different sequences can be used

Tree space may be populated by local optima and islands of optimal trees

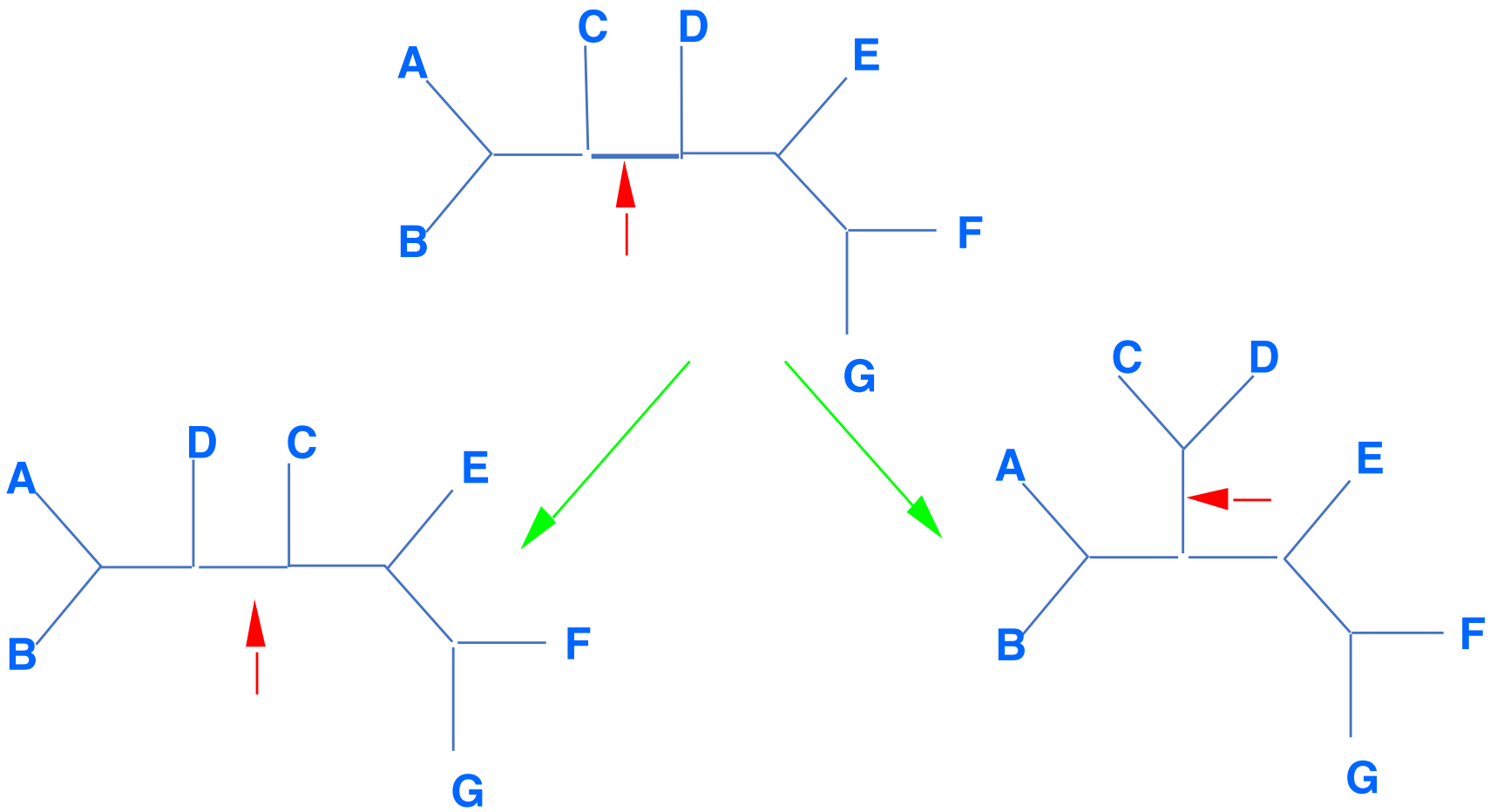


Finding optimal trees – branch swapping

- Nearest neighbor interchange (NNI)
- Subtree pruning and regrafting (SPR)
- Tree bisection and reconnection (TBR)

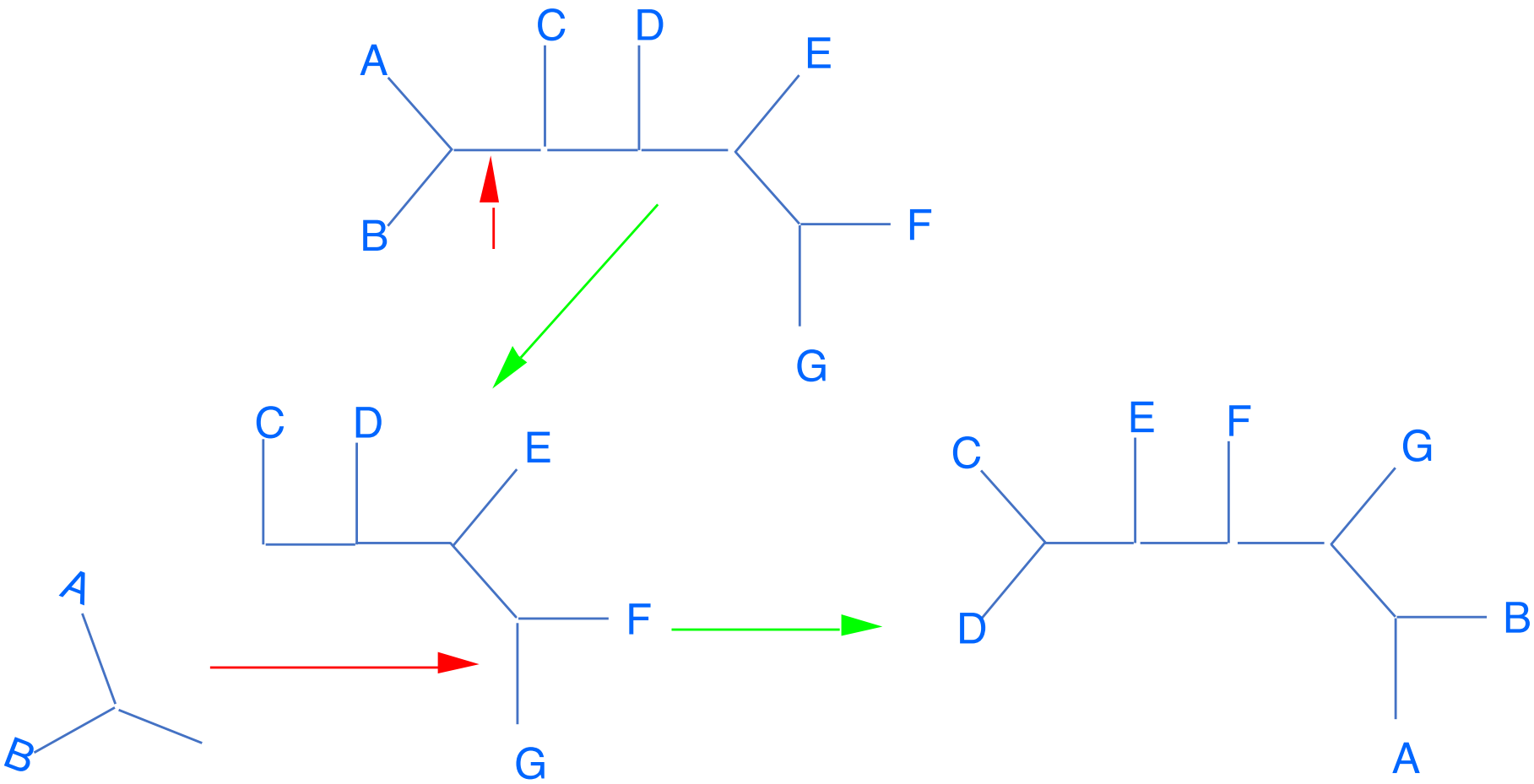
Finding optimal trees - heuristics

Nearest neighbor interchange (NNI)



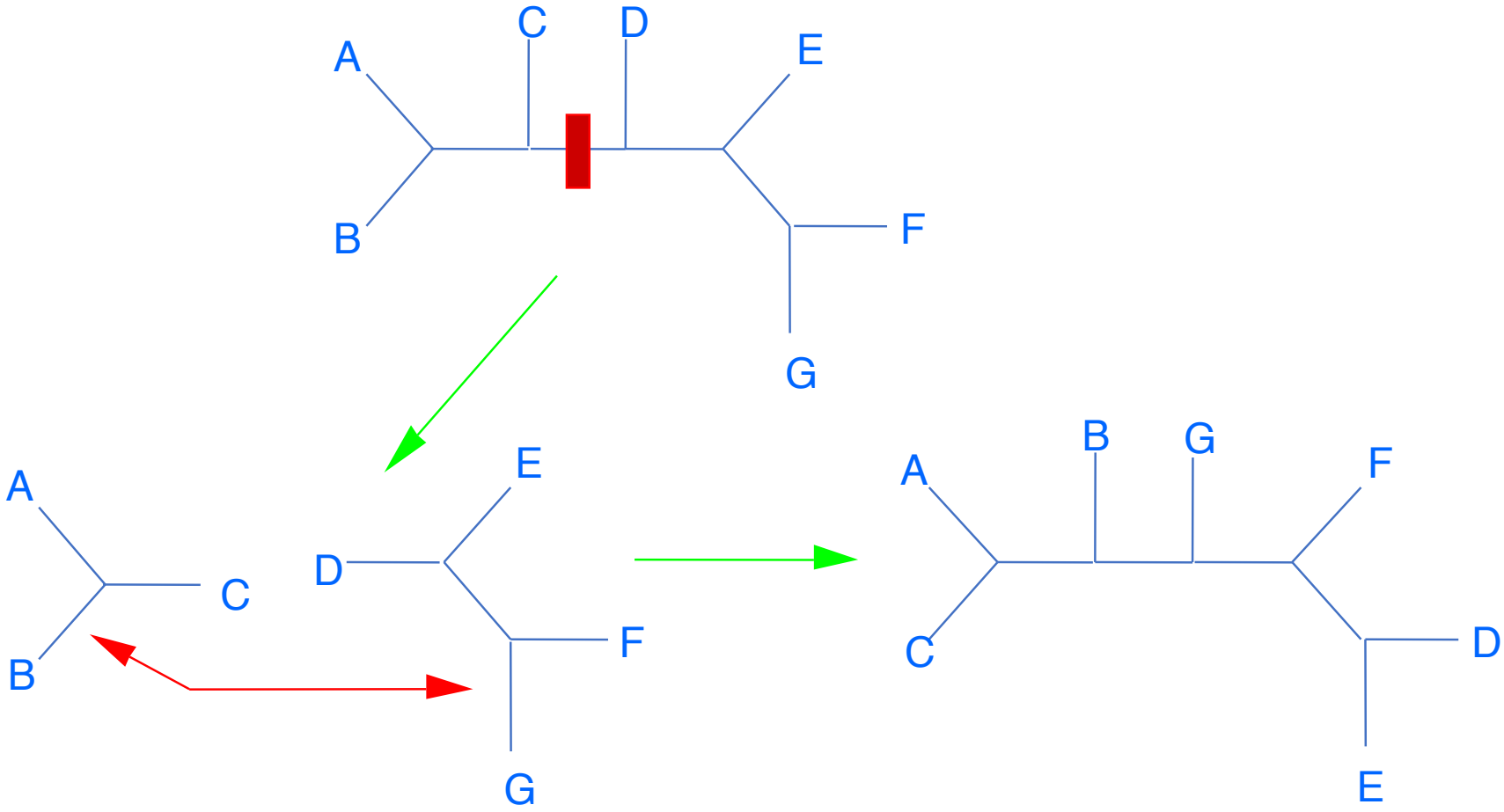
Finding optimal trees - heuristics

Subtree pruning and regrafting (SPR)



Finding optimal trees - heuristics

Tree bisection and reconnection (TBR)

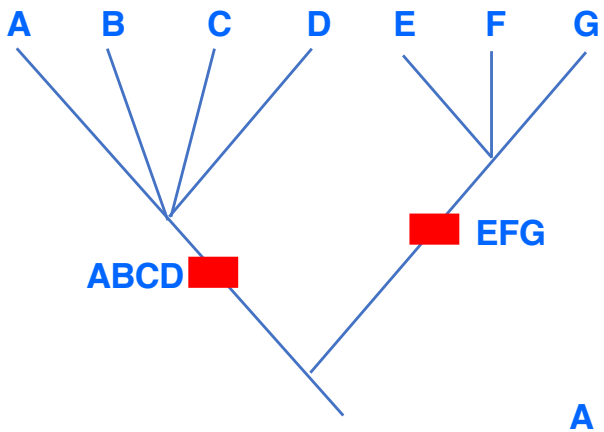


Searching with topological constraints

- Topological constraints are user-defined phylogenetic hypotheses
- Can be used to find optimal trees that either:

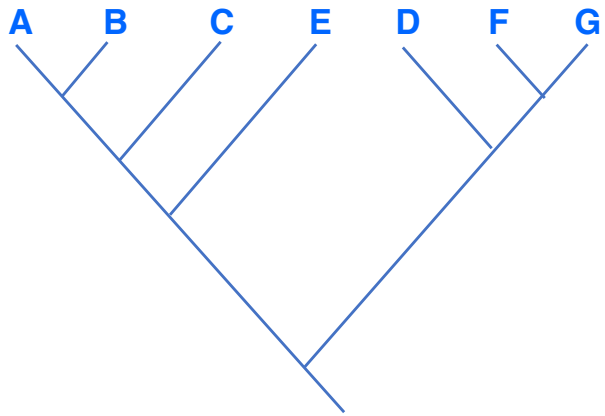
1. include a specified clade or set of relationships
2. exclude a specified clade or set of relationships (reverse constraint)

Searching with topological constraints

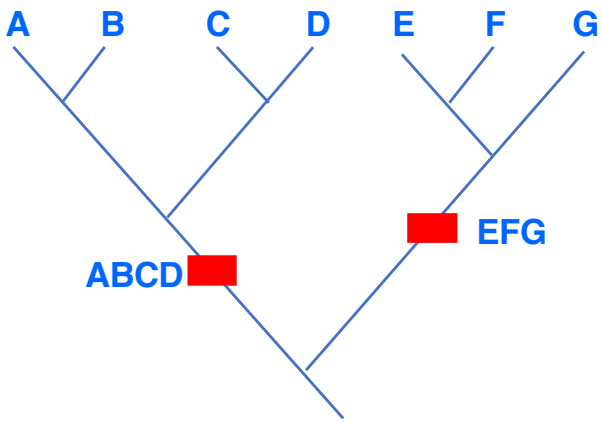


CONSTRAINT TREE

((A,B,C,D)(E,F,G))



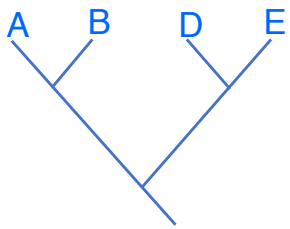
Incompatible with constraint tree



Compatible with constraint tree

Searching with topological constraints
backbone constraints

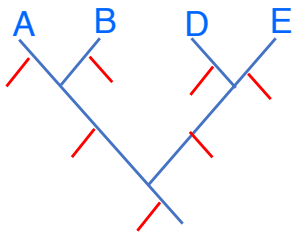
- Backbone constraints specify relationships among a subset of the taxa
- Taxa A, B, C, D, E



BACKBONE CONSTRAINT

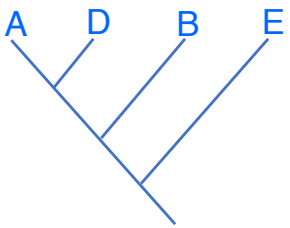
((A,B)(D,E))

relationships of taxon C are not specified



possible positions of taxon C

Compatible with backbone constraint



Incompatible with backbone constraint

Consensus methods

Multiple optimal trees

- Many methods can yield multiple equally optimal trees
- We can further select among these trees with additional criteria, but
- Typically, relationships common to all the optimal trees are summarised with *consensus trees*

Consensus methods

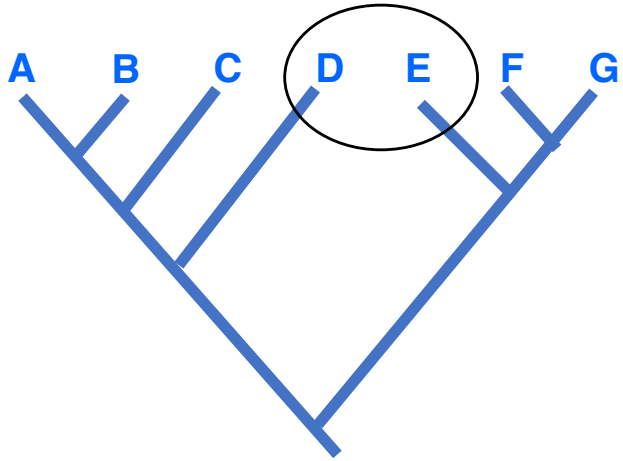
- A consensus tree is a summary of the agreement among a set of fundamental trees
- There are many consensus methods that differ in:
  1. the kind of agreement
  2. the level of agreement
- Consensus methods can be used with multiple trees from a single analysis or from multiple analyses

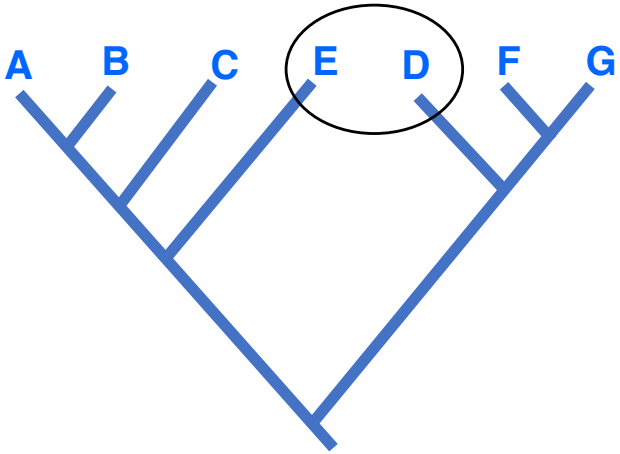
Strict consensus methods

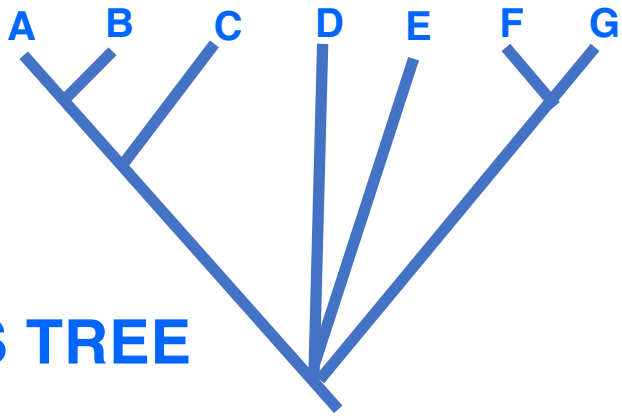
- Strict consensus methods require **agreement across all** the fundamental trees
- They show only those relationships that are **unambiguously supported** by the parsimonious interpretation of the data
- This method produces a consensus tree that includes **all and only those full splits found in all the fundamental trees**
- Other relationships (those in which the fundamental trees disagree) are shown as unresolved polytomies

Strict consensus methods

TWO FUNDAMENTAL TREES







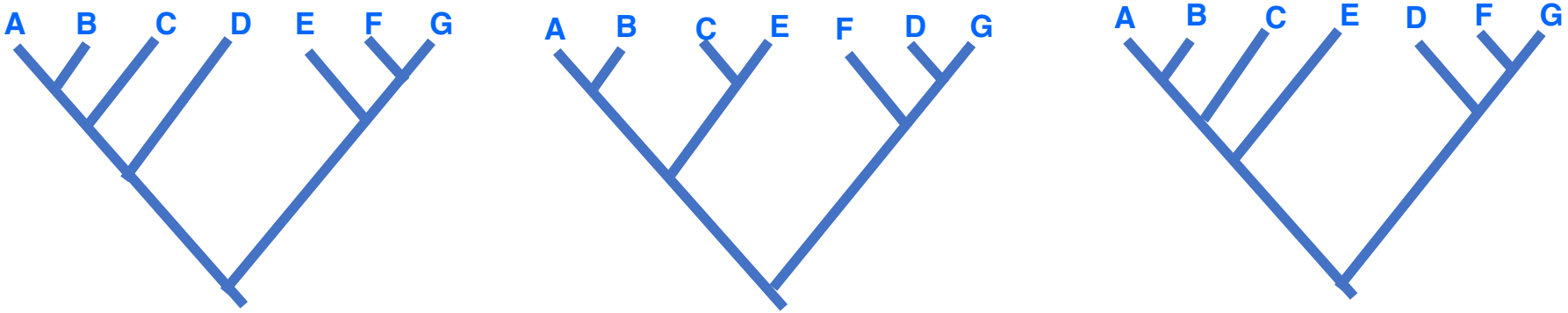
STRICT CONSENSUS TREE

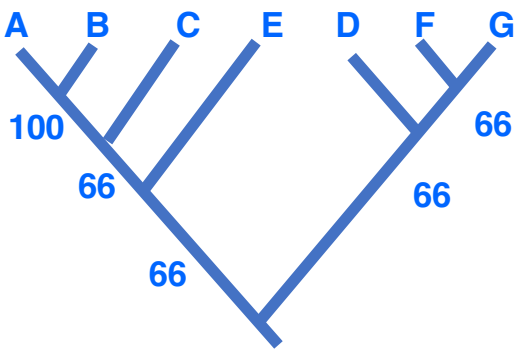
Majority-rule consensus methods

- Majority-rule consensus methods require **agreement across a majority** of the fundamental trees
- May include relationships that are not supported by the most parsimonious interpretation of the data
- This method produces a consensus tree **that includes all and only those full splits found in a majority (>50%) of the fundamental trees**
- Other relationships are shown as unresolved polytomies
- Of particular use in **bootstrapping**

Majority rule consensus

THREE FUNDAMENTAL TREES





Numbers indicate frequency of clades in the fundamental trees

MAJORITY-RULE CONSENSUS TREE

Consensus methods – use

- Currently majority-rule methods mainly used
  - bootstrapping
  - Bayesian methods
- Reduced methods can be useful to identify problem taxa
  - E.g. RogueNaRok
- Strict methods mainly used in parsimony analyses
  - rarely used with molecular data

Take home messages

- Statements of homology are the basis of phylogenetics
- Alignments of molecular sequences are very strong statements of positional homology
- Finding an optimal tree is not a trivial task

The Data

File formats

Computer programs

- Multitude of programs available for free!
- Most have their own input format
- Many are ”black box” programs
- Input files are always simple text files!!!

No good online resource available

http://evolution.gs.washington.edu/phylip/software.html

was an attempt but not updated for a long time

Computer programs - ML

- IQ-TREE (recommended)
- RAxML (recommended)
- PHYML
- GARLI

Computer programs- Bayesian inference

- BEAST (recommended)
- MrBayes (recommended)
- BAMBE
- BayesPhylogenies

Viewing trees

- FigTree (recommended)
- Dendroscope (for large trees >200 taxa)
- R

Three most common data formats

- FASTA
- Phylip
- Nexus

Input format - FASTA

```
>Papilio_glaucus_69_3
GAGaTGGAaGACAaGGTTTCGTCGACCCTGTCCGGCCTCGAGGGCGAACT
>Hamearis84_13
GGaATGGAaGAGAAaGTCTCCACAACCCTCTCCGGACTCGAAGGTGAGCT
>Danaus_plexippus108_21
GAGAtGGAGGAGAaGGTCTCCTCCACCCTCTCAGGTCTCGAAGGTGAACT
>Greta_oto70_9
GGAATGGAaGAGAaGGTCTCCTCGACCCTCTCAGGCCTTGAAGGTGAACT
>Amathusia_phidippus114_17
GGaATGGAaGACAAaGTCTCCTCAaCCCTCTCCGGTCTTGAGGGTGAACT
>Morpho_peleides66_5
GGaATGGAGAGAAAaGTCTCTACTACCCTGTCTGGCCTCGAAGGCGAACT
>BrintesiaB01
GGAATGGAaGACAAaGTCTCGTCCACCCTCTCCGGGCTGGAAGGCGAGCT
>Elymnias_casiphone121_20
GAGAwGGaAGAcaAAGTATCCTCCACCCTCTCTGGTCTTGAAGCTGAACT
>Erebia_oemeEW24_7
gGaATGGAaGACAAaGTCTCCTCGACTCTCTCTGGCCTCGAAGGCGAGCT
```

Input format – PHYLIP

```
9 50
Papilio_gl GAGaTGGAaGACAaGGTTTCGTCGACCCTGTCCGGCCTCGAGGGCGAACT
Hamearis84 GGaATGGAaGAGAAaGTCTCCACAACCCTCTCCGGACTCGAAGGTGAGCT
Danaus_ple GAGAtGGAGGAGAaGGTCTCCTCCACCCTCTCAGGTCTCGAAGGTGAACT
Greta_oto7 GGAATGGAaGAGAaGGTCTCCTCGACCCTCTCAGGCCTTGAAGGTGAACT
Amathusia_ GGaATGGAaGACAAaGTCTCCTCAaCCCTCTCCGGTCTTGAGGGTGAACT
Morpho_pel GGaATGGAGAGAAAaGTCTCTACTACCCTGTCTGGCCTCGAAGGCGAACT
BrintesiaB GGAATGGAaGACAAaGTCTCGTCCACCCTCTCCGGGCTGGAAGGCGAGCT
Elymnias_c GAGAwGGaAGAcaAAGTATCCTCCACCCTCTCTGGTCTTGAAGCTGAACT
Erebia_oem gGaATGGAaGACAAaGTCTCCTCGACTCTCTCTGGCCTCGAAGGCGAGCT
```

Input format - NEXUS

- Nexus blocks: data block, tree block, character set block, MrBayes block, …

```
#NEXUS
BEGIN DATA;
   DIMENSIONS  NTAX=9 NCHAR=50;
   FORMAT DATATYPE=DNA MISSING=? GAP=- INTERLEAVE=No;
   Matrix

[ArgKin 596]
Papilio_glaucus_69_3      GAGaTGGAaGACAaGGTTTCGTCGACCCTGTCCGGCCTCGAGGGCGAACT
Hamearis84_13             GGaATGGAaGAGAAaGTCTCCACAACCCTCTCCGGACTCGAAGGTGAGCT
Danaus_plexippus108_21    GAGAtGGAGGAGAaGGTCTCCTCCACCCTCTCAGGTCTCGAAGGTGAACT
Greta_oto70_9             GGAATGGAaGAGAaGGTCTCCTCGACCCTCTCAGGCCTTGAAGGTGAACT
Amathusia_phidippus114_17 GGaATGGAaGACAAaGTCTCCTCAaCCCTCTCCGGTCTTGAGGGTGAACT
Morpho_peleides66_5       GGaATGGAGAGAAAaGTCTCTACTACCCTGTCTGGCCTCGAAGGCGAACT
BrintesiaB01              GGAATGGAaGACAAaGTCTCGTCCACCCTCTCCGGGCTGGAAGGCGAGCT
Elymnias_casiphone121_20  GAGAwGGaAGAcaAAGTATCCTCCACCCTCTCTGGTCTTGAAGCTGAACT
Erebia_oemeEW24_7         gGaATGGAaGACAAaGTCTCCTCGACTCTCTCTGGCCTCGAAGGCGAGCT
;
end;
```

A good overview available at https://plewis.github.io/nexus/

Input format – NEXUS interleaved

```
#NEXUS
BEGIN DATA;
   DIMENSIONS  NTAX=9 NCHAR=121;
   FORMAT DATATYPE=DNA MISSING=? GAP=- INTERLEAVE=Yes;
   Matrix

[ArgKin 50 bp]
Papilio_glaucus_69_3      GAGaTGGAaGACAaGGTTTCGTCGACCCTGTCCGGCCTCGAGGGCGAACT
Hamearis84_13             GGaATGGAaGAGAAaGTCTCCACAACCCTCTCCGGACTCGAAGGTGAGCT
Danaus_plexippus108_21    GAGAtGGAGGAGAaGGTCTCCTCCACCCTCTCAGGTCTCGAAGGTGAACT
Greta_oto70_9             GGAATGGAaGAGAaGGTCTCCTCGACCCTCTCAGGCCTTGAAGGTGAACT
Amathusia_phidippus114_17 GGaATGGAaGACAAaGTCTCCTCAaCCCTCTCCGGTCTTGAGGGTGAACT
Morpho_peleides66_5       GGaATGGAGAGAAAaGTCTCTACTACCCTGTCTGGCCTCGAAGGCGAACT
BrintesiaB01              GGAATGGAaGACAAaGTCTCGTCCACCCTCTCCGGGCTGGAAGGCGAGCT
Elymnias_casiphone121_20  GAGAwGGaAGAcaAAGTATCCTCCACCCTCTCTGGTCTTGAAGCTGAACT
Erebia_oemeEW24_7         gGaATGGAaGACAAaGTCTCCTCGACTCTCTCTGGCCTCGAAGGCGAGCT

[COI 71 bp]
Papilio_glaucus_69_3      taAagAtaTTgGaACATTATACTTTATTTTTGGAATTTGAGCAAGAATATTAGGAACTTCTTTAAGTTTAT
Hamearis84_13             ??????????????????????????????????????TGAGCAGGAATAGTAGGAACATCATTAAGATTAC
Libythea_celtis71_1       ??????????????????????????????????????TGAGCAGGAATAGTAGGAACTTCATTAAGTCTAT
Danaus_plexippus108_21    ??????????????????????????????????????TGAGCAGGAATAGTTGGGACATCTTTAAGTCTTT
Greta_oto70_9             ??????????????????????????????????????TGAGCAGGAATAGTAGGAACATCTTTAAGTTTAT
Amathusia_phidippus114_17 ??????????????????????????????????????TGATCTGGAATAGTAGGAACATCCCTCAGTCTTA
Morpho_peleides66_5       ??????????????????????????????????????TGAGCCGGTATAATTGGTACATCCCTAAGTCTTA
BrintesiaB01              ??????????????????????????????????????TGAGCAGGTATAGTAGGAACATCTCTTAGTTTAA
Elymnias_casiphone121_20  ??????????????????????????????????????TGATCAGGAATAGTAGGAACTTCCCTCAGTCTTA
Erebia_oemeEW24_7         ??????????????????????????????????????TGAGCAGGTATAGTAGGTACTTCCCTTAGTCTTA
;
end;
```